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31 MARCH 1958
Superseding
MIL-R-19750(Aer)
21 December 1956

MILITARY SPECIFICATION

REGULATOR; VOLTAGE, 28V, 120°C, FOR 30V DIRECT-CURRENT GENERATOR, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers shock mounted, high temperature voltage regulators for use to control 30-volt direct-current aircraft generators.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-T-60	Tape; Pressure Sensitive Adhesive Waterproof - for Packaging and Sealing

Military

JAN-P-100	Packaging and Packing for Overseas Shipment - General Specification
MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-B-4229	Boxes; Paperboard, Metal-Stayed
MIL-D-5028	Drawings and Data Lists; Preparation of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-G-6162	Generators, 30-Volt, Direct Current, Aircraft Engine Driven, General Specification for
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment

FSC 2925

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MIL-W-7139	Wire, Electrical, Insulated, Copper, 600-Volt, 400°Fahrenheit
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-E-7894	Electric Power, Aircraft, Characteristics of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-B-10377	Boxes: Wood-Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, Waterproof

STANDARDS

Military

MIL-STD-15	Electrical and Electronic Symbols
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS25220	Regulator; Voltage, 28 Volt, 120°C, for 30V Direct Current
MS33540	Safety Wiring - General Practices for
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN3632	Generator - Direct Current, 200 Ampere 30 Volt Wide Speed Range
AN3633	Generator - Direct Current, 300 Ampere 30 Volt, Wide Speed Range
AN3634	Generator, Direct Current, 400 Amperes, 30 Volt, Wide Speed Range

PUBLICATIONS

Air Force-Navy Aeronautical Bulletins

No. 143	Specifications and Standards; Use of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The requirements specified in the applicable MS standards are applicable as detail requirements of this specification. Whenever the requirements of the specification and the applicable MS standards conflict, the requirements of the MS standard shall govern.

3.1.1 Applicable MS Standards.- Standard MS25220 is an applicable standard to this specification.

3.1.2 Qualification.- The voltage regulator furnished under this Specification shall be a product which has been tested and has passed the Qualification Tests specified herein.

3.2 Materials.-

3.2.1 Corrosion resistance.- Materials shall be of a corrosion-resisting type or suitably processed to resist corrosion. The following factors shall be used to evaluate the regulators resistance to corrosion; any corrosion that causes:

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(a) Malfunctioning of the equipment.
(b) Shortening of life.
(c) Impairment of use.
(d) Impairment of ease of replacement of parts.

May be deemed sufficient cause for rejection.

3.2.2 Dissimilar metals.- Unless otherwise suitably protected, dissimilar metals, such as brass, copper, or steel, shall not be used in intimate contact with magnesium, aluminum, or their alloys. When protection is used, it shall be of such type that a low-impedance path is offered to radio frequency currents. Dissimilar metals are defined in Standard MS33586.

3.2.3 Finish.-

3.2.3.1 Aluminum alloys.- Aluminum structural parts which do not need to be grounded or bonded shall be anodized in accordance with the requirements of Specification MIL-A-8625 or shall receive an approved chemical film in accordance with the requirements of Specification MIL-C-5541. For aluminum parts or chassis which need to be electrically bonded or grounded, a corrosion-resistant alloy shall be used. This alloy shall be subjected to caustic dip with or without a lacquer finish in lieu of anodizing.

3.2.3.2 Magnesium-alloy parts.- Whenever practicable, magnesium-alloy parts shall be surface-treated in accordance with Specification MIL-M-3171 to provide protection against corrosion.

3.2.3.3 Plating.- Cadmium plating shall be in accordance with Specification QQ-P-416.

3.2.3.4 Electrical sheet steel assemblies.- Electrical sheet steel assemblies, such as armatures and field poles, shall be processed by accepted commercial methods to provide protection against corrosion.

3.2.4 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.4.1 Standard parts.- Standard parts (AN, JAN, MIL, or MS) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part number. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used provided they possess suitable properties and are replaceable by the standard parts without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, when practicable, on the contractor's drawings. In case there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2.4.2 Wire.- The wire used in the voltage regulator shall conform to Specification MIL-W-7139.

3.2.4.3 Screw threads.- Screw threads shall conform to the requirements of Specification MIL-S-7742.

3.2.5 Performance objectives.- Minimum size and weight, simplicity of operation, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives to be considered in the production

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of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the activity responsible for qualification (see 6.3) for consideration. Each request should be accompanied by complete supporting information.

3.3 Design and construction.-

3.3.1 Voltage setting.- The following requirements shall be met with the voltage regulator at room ambient and with a generator conforming to Specification MIL-G-6162 operating at average rated speed and load and rated airflow. The voltage regulator shall be adjusted at room ambient in accordance with the applicable MS Standard except that the final setting of the voltage setting potentiometer shall be at the mechanical midpoint for a regulated output of 28 volts. The voltage setting potentiometer shall be capable of changing the voltage between the limits of 27 +0 -0.5 volts and 29 +0.5 -0 volts. Voltage changes shall be capable of being made in increments no greater than 0.05 volt. The mechanical midpoint setting of the potentiometer shall be nominal and shall be identified by a suitable mark. Any setting of the potentiometer must be able to be accomplished by the use of a screwdriver.

3.3.2 Adjustment.- The adjusting and setting means shall be so designed and constructed that they will not be displaced by vibration, shock or under any of the test conditions. Adjustment procedure will be as specified on the detailed drawing.

3.3.3 Variable resistance.- The regulator shall be designed with a variable resistance as specified on the applicable drawing for control of generator field currents.

3.3.4 Equalizer coil.- The regulator shall be provided with an equalizer coil for obtaining equal division of the load carried by two or more generators operated in parallel.

3.3.5 Electrical connections.- The electrical connections of the voltage regulator shall be as shown on figure 1.

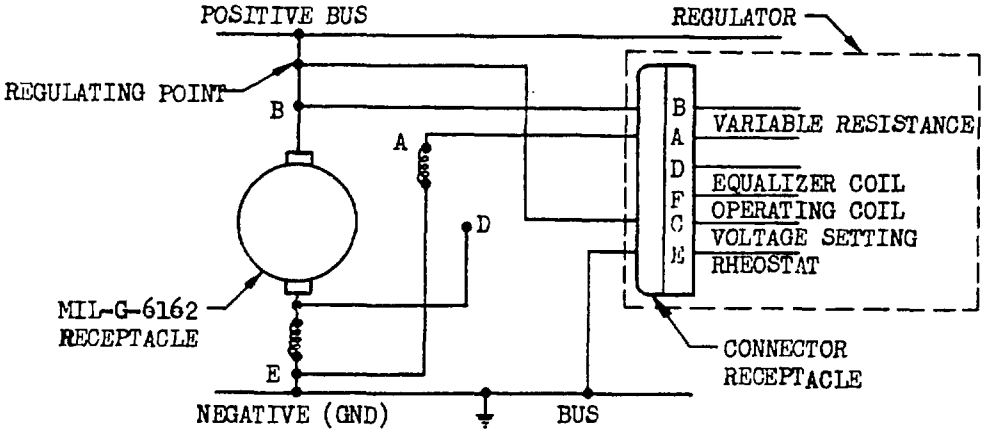


FIGURE 1 - Voltage Regulator Schematic Wiring Diagram

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3.3.6 Connector.- Connections to the regulator will be made through the connector detailed on the applicable drawing.

3.3.7 Ease of maintenance and replacement of parts.- Means shall be provided to remove and replace the carbon pile and remove or recondition both pile buttons without dismantling the unit beyond the removal of the carbon pile adjusting screw. The voltage regulator shall be so designed that all adjustments can be made without dismantling. The voltage regulator shall be so designed that the life of the regulator shall be 2,500 hours or more. The carbon stack life shall be a minimum of 500 hours.

3.3.8 Power requirements.- The total power required for operation of the regulator exclusive of the regulating element dissipation and generator field requirements shall not be more than 40 watts at a maximum of 30 volts direct current. This power shall be applied between pins C and E of the connector with pin E being negative with respect to pin C.

3.3.9 Safety.- Component parts with an electrical potential between the part and ground shall be protected from accidental grounding. The pile adjusting means shall be covered with an insulating cover.

3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Safety wiring and staking.- Accidental loosening of screws and screw parts and other connections shall be prevented by safety wiring, staking, or other methods in conformance with Drawing MS33540.

3.6 Performance.- The voltage regulator shall satisfy the performance requirements specified in Section 4, when subjected to the applicable tests.

3.6.1 General.- Voltage regulators shall conform to the requirements specified in the applicable regulator drawing.

3.6.2 Operating conditions.- The voltage regulator shall be designed for continuous duty under the following individual or natural combinations of operating conditions.

3.6.2.1 Pressure altitude.- Barometric pressure ranging from 30 to 1.68 inches of mercury (approximating an altitude of 65,000 feet). The pressure may remain constant, or may vary at a rate as high as 0.8 inch of mercury per second.

3.6.2.2 Temperature.- Temperature range from -55° to 120°C (see Figure 2).

3.6.2.3 Humidity.- Relative humidity ranging up to 100 percent, including conditions wherein condensation will take place on the equipment.

3.6.2.4 Sand resistance.- Under conditions of airborne sand particles.

3.6.2.5 Resistance to salt spray.- Atmosphere containing salt-laden moisture.

3.6.2.6 Fungus.- As encountered in tropical areas.

3.6.2.7 Acceleration.- Up to values of 10 G's when tested as specified in 4.5.10.

3.6.2.8 Shock.- Up to values of 30 G's when tested as specified in 4.5.11.

3.6.2.9 Vibration.- At rates and amplitudes during the tests specified in 4.5.13.

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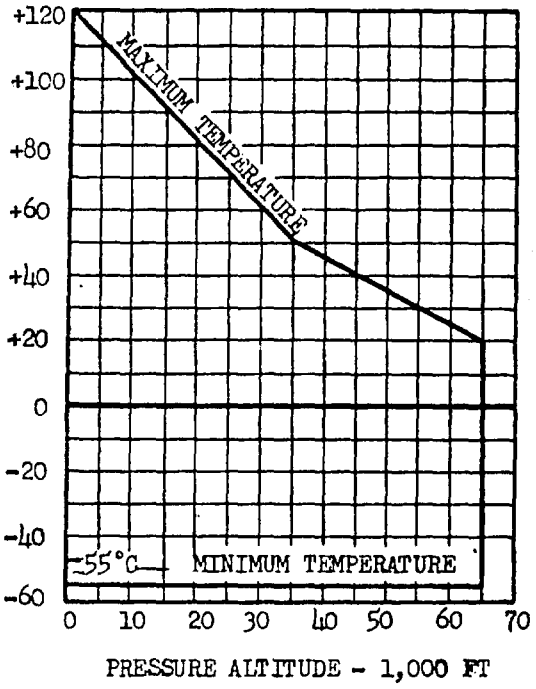


FIGURE 2. Temperature and Altitude Range

3.7 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.7.1 Use of AN or MS designations.- AN or MS designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

3.7.2 Circuit diagram.- Circuit diagrams shall use symbols as specified in Standard MIL-STD-15.

3.8 Installation, testing and maintenance instructions.- The contractor shall pack with each regulator assembly one printed copy of simple instructions with illustrations and diagrams for the purpose of covering the installation of the regulator assembly. The contractor shall also supply tentative but complete test and maintenance instructions with each regulator assembly. Prior to printing, two copies shall be furnished to the Government for approval. Instructions shall be printed on 8-1/2 by 11 inch or 17 by 11 inch paper.

3.9 Workmanship.- All machined surfaces shall have a smooth finish and all details of manufacture, including the preparation of parts and accessories, shall be in accordance with the best practice for high-quality electrical equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, lacquering, riveting, clearance between soldered connections, and ruggedness

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of voltage regulators shall be classified as follows:

(a) Qualification Tests (see 4.2).
(b) Acceptance Tests (see 4.3).

4.2 Qualification Tests.-

4.2.1 Sampling instructions.- Qualification test samples shall consist of six voltage regulators, with the reproducible copies of outline and assembly drawings upon which qualification is desired. Samples shall be forwarded to the agency designated in the letter of authorization from the activity responsible for qualification (see 6.3), plainly identified by securely attached durable tags marked with the following information:

Sample for Qualification Tests
Regulator; Voltage 28-Volt, 120°C, for 30V Direct Current Generator
MS Part No.
Manufacturer's Part No.
Date of Manufacture
Submitted by (name) (date) for Qualification Tests in accordance with the requirements of Specification MIL-R-19750A under authorization (reference authorizing letter).

4.2.2 Tests.- The Qualification Tests shall consist of all the tests of this specification described under Test Methods. These tests may be conducted in any order, unless specifically stated otherwise. However, the more destructive tests will be completed last. Prior to conducting the following tests, each regulator shall pass the tests specified in 4.3.1.

Regulator No. 1

Voltage regulation (entire test)	(4.5.2)
Radio interference	(4.5.9)
Temperature and altitude	(4.5.8)
Shock	(4.5.11)

Regulator No. 2

Voltage regulation (entire test)	(4.5.2)
Temperature and altitude	(4.5.8)
Salt spray	(4.5.15)

Regulator No. 3

Ripple voltage	(4.5.7)
Fungus	(4.5.18)

Regulator No. 4

Time of recovery and stability	(4.5.5)
Life test	(4.5.13)
Time of recovery and stability	(4.5.5)
Load cycling	(4.5.14)
Time of recovery and stability	(4.5.5)

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Regulator No. 5

Time of recovery and stability	(4.5.5)
Acceleration	(4.5.10)
Overload current	(4.5.6)
Sand resistance	(4.5.16)

Regulator No. 6

Time of recovery and stability	(4.5.5)
Acceleration	(4.5.10)
Humidity	(4.5.17)

4.2.2.1 The tests of 4.2.2 may be supplemented by tests under actual service conditions at the option of the activity responsible for qualification.

4.2.2.2 Rejection and retest of qualification samples.- Regulators which have been rejected for any reason during Qualification Tests, may be reworked or have parts replaced to correct defects. Before resubmitting the regulator, full particulars concerning the rejection or reason for return to the manufacturer must be submitted in writing to the testing activity and to the applicable Service. Tests shall not be resumed until this report is received. Where Qualification Tests are conducted under the auspices of the manufacturer, the activity responsible for qualification shall be advised upon failure of the qualification sample and of the action taken by the manufacturer with regard to the failure.

4.2.2.2.1 Disassembly and inspection.- At the conclusion of the Qualification Tests, the voltage regulators shall be disassembled and inspected for excessive wear and defects.

4.3 Acceptance Tests.- The Acceptance Tests shall consist of Individual tests.

4.3.1 Individual tests.- Each voltage regulator shall be subjected to the following tests, as described under Test Methods of this specification:

Examination of product	(4.5.1)
Voltage regulation (at room temperature only)	(4.5.2)
Dielectric strength	(4.5.3)
Equalizer coil	(4.5.4)
Variable resistance (low value only)	(4.5.12)

In addition, the voltage regulator shall be subjected to any other tests specified herein which the Inspector considers necessary to determine conformance with the requirements of this specification.

4.3.3 Rejection and retest.- Regulators which have been rejected may be reworked or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specified approval of the procuring activity.

4.4 Test conditions.- Unless otherwise specified, each test in this section shall be made under the following conditions.

4.4.1 Generator used.- The regulator shall be operated to control a generator which conforms with Specification MIL-G-6162. The minimum speed for regulation shall be determined as specified in 4.4.1.1. The maximum speed for regulation shall be determined in accordance with the procedure of Specification MIL-G-6162.

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When checking for stability of the regulator before and after the Load cycling test (4.5.14), one each of the following wide-speed generators will be used: AN3632, AN3633, and AN3634. These generators may be government-furnished items for this test. The regulator shall meet the system requirements of Specification MIL-E-7894 when operating in conjunction with an electric system and regulating a generator conforming to Specification MIL-G-6162.

4.4.1.1 Minimum speed for regulation.- The minimum speed of a generator will be determined by the following procedure:

(a) With full load applied to the generator and the carbon pile shorted out and a resistance of 1.25 ohms in series with the shunt field, increase the speed until the output voltage is 35 volts. Note the speed of the generator at this point. This shall be minimum speed for regulation.
(b) Disconnect the 1.25 ohms and reconnect the stack: the regulated voltage should be approximately 28 volts.
(c) This procedure for determining minimum speed for regulation shall be used whenever minimum speed for regulation is detailed in this specification. This speed,as determined, may vary from that specified by the generator manufacturer. Any such discrepancy shall be disregarded for purposes of this specification.

4.4.2 Mounting position.- The complete regulator assembly shall be mounted so that the base is down and horizontal.

4.4.3 Auxiliary equipment.- No additional equipment, except that indicated by the manufacturer as necessary to meet the requirements of this specification and included in the weight and dimensions of the applicable drawing, shall be used.

4.4.4 Ambient conditions.-

4.4.4.1 Ambient temperature.- The ambient temperature shall be 25° ±10°C and there shall be no forced ventilation of the regulator or the area immediately surrounding the regulator.

4.4.5 Altitude.- The altitude shall be at approximately sea level.

4.4.6 Warmup.- Warmup of the regulator, when it is specified in any test in this specification, shall be accomplished in the following manner:

(a) The regulator shall be connected to control a generator at or above minimum speed for regulation with a load on the generator requiring the variable resistance to be dissipating at least 50 percent of the rated wattage specified on the applicable drawing. The regulator shall operate in this circuit for a period of at least 30 minutes.

4.4.7 Test voltage setting.- The regulated voltage shall be set at 28 volts ±0.1 volts at continuous operating speed at no load on the generator. This value of voltage shall be set prior to starting each test in this specification.

4.4.8 Voltage measurement.-

4.4.8.1 Regulated voltage.- The regulated voltage shall be measured across the operating coil connector pins C and E. Whenever the regulated voltage is to be measured during any of the tests specified herein, it shall be accomplished by means of a laboratory-type voltmeter; in addition, a suitable continuous recording voltmeter shall be used during qualification testing, with the laboratory-type meter serving as a calibration check.

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4.4.8.2 Variable resistance voltage.- The variable resistance voltage shall be measured across the carbon pile connector pins A and B.

4.5. Test Methods.-

4.5.1 Examination of product.- Each regulator, subassembly, and part shall be examined as the Inspector may deem necessary to determine conformance with this specification and the applicable drawings with respect to materials, voltage setting, electrical connections, threaded parts, finish, interchangeability, marking, instructions, dimensions, weight, and workmanship.

4.5.2 Voltage regulation (effect of speed and load).- The regulator shall be warmed up and set as described in 4.4.6 and 4.4.7 prior to starting this test. The regulated voltage shall be measured under the following conditions:

(a) At minimum speed for regulation, the load shall be varied from no load to full load. The load shall be suddenly removed and applied for each observation. (Shock loaded.)
(b) At average rated generator speed, the preceding test shall be repeated.
(c) At maximum rated generator speed, the preceding test shall be repeated.
(d) At zero generator load current, the speed shall be varied from minimum speed for regulation to maximum speed for regulation and returned. The rate of speed change shall be approximately 500 rpm (or less) per second. Do not shock load or bump regulator in any manner.
(e) At 50 percent rated generator load current, the speed shall be varied from minimum speed for regulation to maximum rated speed and returned. Do not shock load or bump regulator in any manner.
(f) At 100 percent rated generator load current, the preceding test shall be repeated. The rate of speed change shall be approximately 500 rpm (or less) per second.

For the entire test the voltage shall remain within the limits of the applicable MS Standard.

4.5.3 Dielectric strength.- For Qualification Tests, while the regulator is hot as a result of testing, a voltage of 1,250 volts rms,60 cycles, shall be applied between independent circuits and between each circuit and frame for 1 minute. The insulation shall not fail during these tests. For production testing, this test shall be conducted by applying 1,250 volts rms, 60 cycles, for 1 second.

4.5.4 Equalizer coil.- The effect of the equalizer coil shall be determined when the regulator is controlling a generator in accordance with Specification MIL-G-6162 and dissipating 50 percent of the minimum wattage specified on the applicable drawing for 30 minutes. The regulator shall be provided with an equalizer coil for obtaining equal division of the load carried by two or more generators operating in parallel. This coil shall lower the controlled voltage by 2.5 ±0.25 volts when 0.25 volt is applied between connector pin F and connector pin D, with pin D negative to pin F. The resistance of this coil, when referred to 20°C shall be 0.4 ±0.04 ohm.

4.5.5 Time of recovery and stability.-

(a) The generator shall be operated at minimum, maximum, and average speeds. A recording oscillograph shall be used to measure the regulated voltage transient when the full load is applied and removed.

(b) The transient voltage must remain within the appropriate limits of Specification MIL-E-7894 when the load is applied and removed. The oscillations shall not cross the value of the steady state voltage more than five times in 0.5 second in returning to within ±0.5 volt of its initial voltage.

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(c) The stability of a regulator shall be determined by the use of a calibrated oscilloscope connected across the variable resistance. The generator shall be operated at a speed which will require the greatest ohmic variation in the variable resistance when full rated load is applied and removed. Continued fluctuation in the level of the controlled voltage beyond the limits of item 8 of the applicable drawing when full load is applied and removed from the generator shall be cause for rejection.

4.5.6 Overload current.- The voltage coil of the voltage regulator shall be disconnected and the overload field current shall be passed through the field circuit for a period of 1 hour. Immediately following the test, the regulator shall be examined. Damage to any of the operating parts shall be cause for rejection.

4.5.7 Ripple voltage.-

(a) Ripple voltage shall be determined by means of a General Radio 727A vacuum tube voltmeter, or equivalent, with a 4.0-microfarad capacitor connected as shown in Figure 3. (This is a peak reading meter but the scale is calibrated in rms for a sine wave, i. e., peak voltage equals scale reading times 1.414.)

(b) The generator shall be operated at maximum rated speed and with no load and full load applied to the generator without a battery in parallel.

(c) Peak voltage readings shall be taken with the voltmeter successively connected for each of the two polarities. A generator shall be used which meets the requirements of Specification MIL-G-6162. A set of readings shall be taken with the voltage regulator, then with the manually operated rheostat connected as shown. The rheostat shall be adjusted to the same operating voltage condition as the regulator. The higher of the two readings taken with the voltage regulator shall not exceed the higher of the readings taken with the rheostats by more than 10 percent.

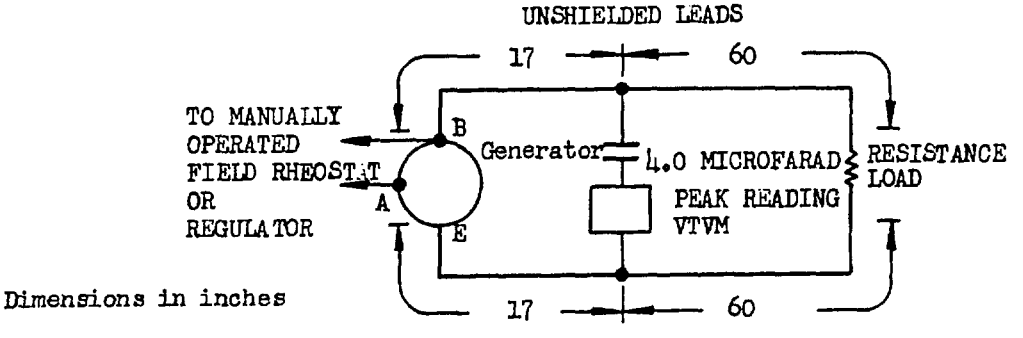


FIGURE 3. Ripple Voltage Test Set-up

4.5.8 Temperature and altitude test.- The complete voltage regulator assembly shall be subjected to temperature and altitude by the procedure described herein:

(a) The regulator shall be instrumented with suitable thermocouples and a recording indicator to record the temperature of the following external points:

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(1) Between two of the central fins and adjacent to the stack tube.
(2) An external edge of a central fin.
(3) At the pile screw end of the housing.
(4) On the magnet case.

This information is to be used only as an indication of what effect the environmental conditions have on parts of the regulator.

(b) Place the complete regulator assembly connected to control a generator of the type as specified by Specification MIL-G-6162 in a temperature and altitude chamber.

(c) Then, the program and requirements shown in Table I shall be met.

4.5.9 Radio interference.- Conducted and radiated radio interference shall be measured in accordance with Specification MIL-I-6181 under the following conditions:

(a) The regulator shall be controlling a generator in accordance with Specification MIL-G-6162 at maximum speed and with full load and no load on the generator.

(b) The conducted radio interference shall be measured between all combinations of two regulator connector pins or regulator connector pins and ground. The value of this conducted radio interference shall not exceed the limits of the conducted radio interference as determined for the generator as specified in the radio interference test of Specification MIL-I-6181.

4.5.10 Acceleration.- The regulator shall be mounted on a centrifuge with the mounting base down and horizontal. The major axis of the carbon pile shall intersect the axis of rotation of the centrifuge. The centrifuge shall then be operated at a speed that will produce an acceleration of 4.0 gravitational units along the major axis of the carbon pile for a period of 30 seconds. Rotate the regulator 180 degrees about the vertical axis of the regulator and repeat the above test.

The regulator shall then be rotated 90 degrees about the vertical axis. The centrifuge will then be operated at a speed that will produce 10 gravitational units along an axis perpendicular to the major axis of the carbon pile for a period of 1 minute. Rotate the regulator 180 degrees about the vertical axis of the regulator and repeat the above test.

The deviation in controlled voltage produced by the acceleration shall be measured when the regulator is controlling a generator conforming to Specification MIL-G-6162 with the generator operating at a speed and load causing the variable resistance to equal 25 ohms.

The controlled voltage, while the regulator is undergoing acceleration in any of the above directions shall remain between the limits of item 8 of the applicable drawing. The regulator shall show no sign of instability or continued arcing in the variable resistance.

4.5.11 Shock.- The complete regulator assembly nonoperating shall be subjected to the following Shock Test:

(a) Test conditions shall be as follows:

(1) 18 impact shocks of 15 ±1 gravitational units
(2) Three shocks in each direction along each of three mutually perpendicular axes.

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(3) The time duration of shock shall be 11 ±1 milliseconds and the maximum gravity shall occur at approximately 5-1/2 milliseconds.

(b) The complete regulator assembly shall pass this test and the voltage regulation tests (4.5.2) without readjustment.

(c) The structural integrity of the mounting base and isolators shall be determined with either the regulator or a dummy load in place when subjected to the following test:

(1) With excursion stops or bumpers in place, the mounting base individual isolators, or other attaching devices, the shock shall consist of at least 12 impacts of 30 gravitational units.
(2) The shocks, two each, in opposite directions along each of three mutually perpendicular major axes of the regulator.
(3) Time duration 11 ±1 milliseconds and the maximum gravity shall occur at approximately 5-1/2 milliseconds.

(d) Bending and distortion shall be permitted; however, there shall be no failure of the attaching joints, and the regulator or dummy load shall remain in place.

4.5.12 Variable resistance - Low and High values.- This test shall be made immediately following satisfactory completion of the voltage regulation test at room ambient:

(a) The regulator shall be operated to control a generator at or above minimum speed for regulation and a load shall be applied requiring the regulator variable resistance to be dissipating 50 ±5 watts.

(b) After regulator temperatures have stabilized, the rated load generator current shall be switched off and on and the regulated voltage at generator load current shall then be read.

(1) Full load - low value.- The rated generator load current shall be switched off and on continuously (approximately 3 seconds on and 2 seconds off cycling) while the generator speed is lowered. When the regulated voltage at the rated generator load current is 26.5 volts, the variable resistance current and voltage shall be read. The low value of variable resistance shall be the quotient of the variable resistance voltage divided by the variable resistance current. This value shall be no greater than the value specified on the applicable MS standard.

(2) No load - High value.- Operate the generator at no load and its maximum speed for regulation. Measure the regulator value of variable resistance. The resistance shall meet the high value requirement specified by the applicable MS standard.

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TABLE I

TEMPERATURE ALTITUDE

Interval (min)	Temperature (°C)	Altitude (ft)	Voltage Limits (V)	Wattage Dissipation(%)	Remarks
	25 ±10	Sea level	28 ±0.1	50 ±5	Reference run See Notes 1 & 2
	-55	Sea level	Nonoperating	--	See Note 2
0.0 to 0.5	-55	Sea level	26.0 to 30.0	50 ±5	See Note 3
0.5 to 5.0	-55	Sea level	27.5 to 28.5	50 ±5	See Notes 4 & 5
5.0 to 30.0	-55	Increase altitude	27.0 to 29.0	50 ±5	--
30.0 to 60.0	-55	65,000	27.5 to 28.5	50 ±5	See Notes 4 & 5
60.0 to 90.0	Increase temp	65,000	27.0 to 29.0	50 ±5	--
90.0 to 1590	+20	65,000	27.5 to 28.5	Max rated watts	See Notes 4 & 5
1590 to 1620	Increase temp	Decrease altitude	27.0 to 29.0	Max rated watts	See Note 5
1620 to 1650	+50	35,000	27.5 to 28.5	Max rated watts	See Notes 4 & 5
1650 to 1680	Increase temp	Decrease altitude	27.0 to 29.0	Max rated watts	--
1680 to 1780	+120	Sea level	27.5 to 28.5	Max rated watts	See Notes 4 & 5
1780 to 1810	Decrease	Sea level	27.0 to 29.0	Max rated watts	--
1810 to 1910	25 ±10	Sea level	27.5 to 28.5	50 ±5	See Note 4

NOTES:

1. The regulated voltage shall be set at 28 volts as described in 4.4.7.
2. With the regulator deenergized, reduce the chamber temperature to -55°C at sea level and soak the regulator at this temperature for a minimum period of 12 hours.
3. After a minimum of 12 hours cold soaking and with the regulator deenergized, start the generator and increase the speed to the minimum rated speed as determined in 4.4.1.1 and then energize the regulator.
4. The regulator shall be subjected to the following regulation test at the conclusion of the time intervals where a regulation test is specified:

   (a) With the regulator controlling the generator, operate the generator at minimum speed for regulation (as determined in 4.4.1.1) at no load, half-rated, and rated loads.
   (b) Increase the speed of the generator to average rated speed and repeat the load conditions as in subparagraph 4.(a) above.
   (c) Increase the speed of the generator to maximum rated speed and repeat the load conditions as in subparagraph 4.(a) above.
   (d) The controlled voltage shall remain within the limits of this table.

5. When maximum rated wattage dissipation is specified in this table, the conditions may be synthesized to obtain the desired dissipation but must be removed to perform the voltage regulation tests.

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4.5.13 Life test.- The life test of the complete regulator assembly need not be a continuous life test but must be consecutive.

(a) Mount the regulator on a vibration table. The mounting shall be with the base horizontal and down. Then the procedures listed shall be followed:

(1) Vibration Limits.- With the regulator operating and controlling a generator conforming to Specification MIL-G-6162, vibrate the table in a horizontal direction with the frequency varying between 10 to 500 cycles per second. From 10 to 60 cycles per second the excursion shall be 0.03 inch (double excursion 0.06 inch). From 60 to 500 cycles per second the applied vibratory acceleration shall be 10 gravity units. The rate of change of the vibration frequency shall be such that a complete cycle will consume approximately 15 minutes. This test shall continue for a period of 240 hours. The time shall be equally divided and the vibration shall be applied along each of three mutually perpendicular axes.

(2) Ohmic variation.- The following test conditions shall be applied during the 240 hour life test of (1) above. The value of the variable resistance of the regulator shall be 15 ohms for 58 minutes, and for 2 minutes the condition shall be such that the variable resistance shall increase to 25 ohms. Every tenth hour the circuit conditions shall be such that the variable resistance shall increase to 35 ohms for a maximum period of 2 minutes. Special circuitry will be permitted to enable the regulator to obtain the maximum ohmic variation.

(b) Performance.- During any 120-hour period of testing, the regulated voltage shall not have exceeded the tolerance limits of item 7 of the applicable drawing at a base voltage of 28 ±0 volts.

4.5.14 Load cycling test.- Immediately at the conclusion of the 240 hours of life test, conduct a check for stability of the regulator as described in 4.5.5(c). If the regulator is stable, proceed with the following endurance test for 150 hours:

(a) With regulator connected and controlling a generator conforming to Specification MIL-G-6162, the generator shall be operated at a speed which will require the greatest ohmic variation in the variable resistance when full rated load is switched off and on the generator. The full rated load shall be switched on for 5 seconds and off for 5 seconds for the entire 150 hours.

(b) An oscilloscope shall be connected across the variable resistance to detect any instability during the course of this endurance test. The controlled voltage shall remain within the limits of item 8 of the applicable drawing.

(c) At the conclusion of the 150-hour load cycling test the regulator shall pass the stability test of paragraph 4.5.5(c) and the voltage regulation test at room ambient and the controlled voltage shall remain within the limits of item 8 with the tolerance limits specified on the applicable MS standard.

4.5.15 Salt spray.- The regulator nonoperating shall be subjected to the Salt spray test described in Specification MIL-E-5272 except it shall be 25 continuous hours. At the conclusion of the salt spray test, the regulator may be water washed of any salt deposit and the carbon stack and buttons may be replaced. The voltage regulator may be

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readjusted and reset in accordance with 3.3.4 and 3.3.5. The regulator shall meet the voltage regulation test (see 4.5.2) at room ambient and the regulated voltage shall remain within the limits of item 8 of the applicable drawing. Any condition of corrosion that results in effects described in 3.2.1 shall be considered cause for rejection.

4.5.16 Sand resistance.- The regulator shall be subjected to the sand and dust test of Specification MIL-E-5272, procedure II, and the following operating conditions shall apply:

(a) The regulator shall be tested for a total of 8 hours. The test cycle shall be composed of 30 minutes operative and 90 minutes inoperative periods. The total operative test duration shall be 2 hours and the inoperative test duration shall be 6 hours.

(b) During the operative periods, the regulator shall be controlling a generator of the type specified in Specification MIL-G-6162 operating at average rated speed with no load applied for the first 10 minutes and full load applied for the next 20 minutes.

(c) At the conclusion of the test, the regulator shall pass the voltage regulation test (see 4.5.2) at room ambient without resetting and the controlled voltage shall be within the limits of item 8 and with tolerance limits of item 7 of the applicable drawing.

4.5.17 Humidity.- The regulator, nonoperating, shall be subjected to the humidity test specified in Specification MIL-E-5272, procedure I, except the cycle shall be repeated a sufficient number of times to extend the total time of the test to 240 hours (10 cycles). At the conclusion of the test the regulator shall pass the voltage regulation test (see 4.5.2) at room ambient without resetting and the controlled voltage shall be within the limits of item ' of the applicable drawing. Any condition of corrosion that results in effects described in paragraph 3.2.1 of this specification shall be considered cause for rejection.

4.5.18 Fungus tests.- The relative humidity for this test shall be 95 ±5 percent. The temperature for this test shall be 30° ±3°C. The regulator shall be subjected to one fungus from each of the following groups:

Group	
Group I	Chaetomium globosum USDA 1042.4 or Myrothecium verrucaria USDA 1334.2.
Group II	Rhizopus nigricans S. N. 32 or Aspergillus niger USDA Tc215-4247.
Group III	Aspergillus flavus AMC No. 26 or Aspergillus terreus PQMD82j.
Group IV	Penicillium luteum USDA 1336.1 Penicillium sp. USDA 1336.2 or Penicillium citrinum ATCC 9848.
Group V	Mennoniella echinata AMC No. 37 or Fusarium moniliforme USDA 1004.1.

Substitutes for any of the above fungi may be made if specified in the detail specification. The regulator will be either dipped in or sprayed with a spore suspension before exposure to test condition. Duration of test shall be 28 days. Following the test, the regulator shall be inspected for general deterioration and shall meet the voltage regulation test at room ambient without readjustment or resetting.

4.5.19 Explosion proof.- The voltage regulator shall be subjected to the Explosion proof test, Procedure II, of Specification MIL-E-5272. The regulator shall meet the test requirements.

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4.6 Material test specimens.- Test specimens shall be furnished when requested by the Government for check tests at an acceptable Government laboratory to determine conformance with the applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Regulators shall be preserved in accordance with MIL-P-116, Method II, and packed in containers meeting Specification MIL-B-4229, PPP-B-566, PPP-B-676, or PPP-B-636. For domestic shipment, containers meeting Specification PPP-B-636 may be used. Unit quantities shall be as specified by the procuring activity.

5.1.2 Level C.- Regulators shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.-

5.2.1 Level A.- Regulators preserved and packaged to meet 5.1.1 shall be packed in exterior type shipping containers meeting Specification PPP-B-585, PPP-B-591, PPP-B-621, PPP-B-601, PPP-B-636, or MIL-B-10377. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to Specification MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to Specification PPP-B-636 and is sealed at all joints and seams, including the manufacturer's joint, with tape conforming to Specification PPP-T-60.

5.2.2 Level B.- Regulators preserved and packaged to meet 5.1.1, or 5.1.2 shall be packed in domestic type exterior containers meeting Specification PPP-B-591, PPP-B-585, PPP-B-621, PPP-B-601, PPP-B-636, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall meet the requirements of table I of Specification PPP-B-636.

5.2.3 Level C.- Packages which require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportaion.

5.3 Physical protection.- Cushioning, blocking, bracing, and bolting as required shall be in accordance with Specification JAN-P-100 except that for domestic shipments, water-proofing requirements for cushioning materials and containers shall be waived. Drop tests of Specification JAN-P-100 shall be waived when preservation, packaging, and packing of the item are for immediate use or when drop tests of Specification MIL-P-116 are applicable.

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5.4 Marking for shipment.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The following additional information shall be included:

MS Part No.

If in storage after (date)* the regulator shall be tested and inspected before issue.

*This date shall be 36 months from date of inspection, and shall be inserted by the inspector.

6. NOTES

6.1 Intended use.- This voltage regulator is intended to control a 30-volt direct current aircraft generator in a 28-volt dc (nominal) aircraft electrical system.

6.2 Ordering data.- Requisitions, contracts, and orders should include the MS part number and whether overseas packing is desired. (See section 5.)

6.3 Provisions for qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification should be addressed to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., the activity responsible for qualification, with a copy to the Commander, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio.

6.4 Definition.-

6.4.1 Voltage setting.- Voltage setting shall mean the voltage obtained by use of the voltage setting potentiometer.

6.4.2 Adjustment.- Adjustment as used in this specification shall refer to the procedures used in adjusting the pile screw and the core adjustment so that the regulator is operating at the proper point on the curve of voltage versus pile screw positions.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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